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User-assisted robotic control systems

Abstract

Exemplary embodiments relate to user-assisted robotic control systems, user interfaces for remote control of robotic systems, vision systems in robotic control systems, and modular grippers for use by robotic systems. The systems, methods, apparatuses and computer-readable media instructions described interact with and control robotic systems, in particular pick and place systems using soft robotic actuators to grasp, move and release target objects.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/137,926, filed on Apr. 21, 2023, which is a continuation of U.S. patent application Ser. No. 17/495,125, filed on Oct. 6, 2021, which is a continuation of, claims the benefit of and priority to previously filed U.S. patent application Ser. No. 15/941,670 filed on Mar. 30, 2018, entitled “User-Assisted Robotic Control Systems”, which claims priority to U.S. Provisional Application Ser. No. 62/478,775, filed on Mar. 30, 2017, entitled “User-Assisted Robotic Control Systems.” The aforementioned applications are incorporated herein by reference in their entireties.

FIELD OF THE DISCLOSURE

(1) The disclosure relates generally to the field of robotics and particularly to novel systems for controlling a robotic system. Exemplary embodiments may be employed in a variety of applications, including (but not limited to) warehouse management, order fulfillment, packaging, recycling, etc.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A-D depict exemplary soft robotic actuators that may be used according to exemplary embodiments;
- (2) FIG. 2A-G depict an exemplary robotic system architecture according to exemplary embodiments;
- (3) FIG. 3A-H depict a user interface according to exemplary embodiments;
- (4) FIG. 4A-D depict calibration of an exemplary system according to exemplary embodiments;
- (5) FIG. 5A-D depict a portion of the exemplary servo-pneumatic control system in unactuated and actuated states, respectively, according to exemplary embodiments;
- (6) FIG. 6A-P depict components and views of a modular gripper according to exemplary embodiments;
- (7) FIG. 7 is a block diagram illustrating an exemplary computing device suitable for use with exemplary embodiments; and
- (8) FIG. 8 depicts an exemplary communication architecture.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(9) The present invention will now be described more with reference to the accompanying drawings, in which preferred embodiments of the invention are shown. The invention, however, may be embodied in many different forms and should not be construed as being limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art. In the drawings, like numbers refer to like elements throughout.

I. INTRODUCTION TO SOFT ROBOTIC GRIPPERS

- (10) Embodiments described herein may be used in connection with hard robotic grippers (e.g., grippers made of a relatively hard or relatively non-deformable material, such as metal, hard plastic, etc.). In many cases, however, soft robotic grippers may provide improvements over hard robotic grippers. The description below provides background information on soft robotics.
- (11) Conventional robotic grippers or actuators may be expensive and incapable of operating in certain environments where the uncertainty and variety in the weight, size and shape of the object

being handled has prevented automated solutions from working in the past. The present application describes applications of novel soft robotic actuators that are adaptive, inexpensive, lightweight, customizable, and simple to use.

(12) Soft robotic actuators may be formed of elastomeric materials, such as rubber, or thin walls of plastic arranged in an accordion structure that is configured to unfold, stretch, twist, and/or bend under pressure, or other suitable relatively soft materials. They may be created, for example, by molding one or more pieces of the elastomeric material into a desired shape. Soft robotic actuators may include a hollow interior that can be filled with an inflation fluid, such as air, water, or saline to pressurize, inflate, and/or actuate the actuator. Upon actuation, the shape or profile of the actuator changes. In the case of an accordion-style actuator (described in more detail below), actuation may cause the actuator to curve or straighten into a predetermined target shape. One or more intermediate target shapes between a fully unactuated shape and a fully actuated shape may be achieved by partially inflating the actuator. Alternatively, or in addition, the actuator may be actuated using a vacuum to remove inflation fluid from the actuator and thereby change the degree to which the actuator bends, twists, and/or extends.

(13) Actuation may also allow the actuator to exert a force on an object, such as an object being grasped or pushed. However, unlike traditional hard robotic actuators, soft actuators maintain adaptive properties when actuated such that the soft actuator can partially or fully conform to the shape of the object being grasped. They can also deflect upon collision with an object, which may be particularly relevant when picking an object off of a pile or out of a bin, since the actuator is likely to collide with neighboring objects in the pile that are not the grasp target, or the sides of the bin. Furthermore, the amount of force applied can be spread out over a larger surface area in a controlled manner because the material can easily deform. In this way, soft robotic actuators can grip objects without damaging them.

(14) Moreover, soft robotic actuators allow for types of motions or combinations of motions (including bending, twisting, extending, and contracting) that can be difficult to achieve with traditional hard robotic actuators.

(15) FIGS. 1A-1D depict exemplary soft robotic actuators. More specifically, FIG. 1A depicts a side view of a portion of a soft robotic actuator. FIG. 1B depicts the portion from FIG. 1A from the top. FIG. 1C depicts a side view of a portion of the soft robotic actuator including a pump that may be manipulated by a user. FIG. 1D depicts an alternative embodiment for the portion depicted in FIG. 1C.

(16) An actuator may be a soft robotic actuator **100**, as depicted in FIG. 1A, which is inflatable with an inflation fluid such as air, water, or saline. The inflation fluid may be provided via an inflation device **120** through a fluidic connection **118**.

(17) The actuator **100** may be in an uninflated state in which a limited amount of inflation fluid is present in the actuator **100** at substantially the same pressure as the ambient environment. The actuator **100** may also be in a fully inflated state in which a predetermined amount of inflation fluid is present in the actuator **100** (the predetermined amount corresponding to a predetermined maximum force to be applied by the actuator **100** or a predetermined maximum pressure applied by the inflation fluid on the actuator **100**). The actuator **100** may also be in a full vacuum state, in which all fluid is removed from the actuator **100**, or a partial vacuum state, in which some fluid is present in the actuator **100** but at a pressure that is less than the ambient pressure. Furthermore, the actuator **100** may be in a partially inflated state in which the actuator **100** contains less than the predetermined amount of inflation fluid that is present in the fully inflated state, but more than no (or very limited) inflation fluid.

(18) In the inflated state, the actuator **100** may exhibit a tendency to curve around a central axis as shown in FIG. 1A. For ease of discussion, several directions are defined herein. An axial direction passes through the central axis around which the actuator **100** curves, as shown in FIG. 1B. A radial direction extends in a direction perpendicular to the axial direction, in the direction of the radius of

the partial circle formed by the inflated actuator **100**. A circumferential direction extends along a circumference of the inflated actuator **100**.

(19) In the inflated state, the actuator **100** may exert a force in the radial direction along the inner circumferential edge of the actuator **100**. For example, the inner side of the distal tip of the actuator **100** exerts a force inward, toward the central axis, which may be leveraged to allow the actuator **100** to grasp an object (potentially in conjunction with one or more additional actuators **100**). The soft robotic actuator **100** may remain relatively conformal when inflated, due to the materials used and the general construction of the actuator **100**.

(20) The actuator **100** may be made of one or more elastomeric materials that allow for a relatively soft or conformal construction. Depending on the application, the elastomeric materials may be selected from a group of food-safe, biocompatible, or medically safe, FDA-approved materials. The actuator **100** may be manufactured in a Good Manufacturing Process (“GMP”)-capable facility.

(21) The actuator **100** may include a base **102** that is substantially flat (although various amendments or appendages may be added to the base **102** in order to improve the actuator's gripping and/or bending capabilities). The base **102** may form a gripping surface that grasps a target object.

(22) The actuator **100** may include one or more accordion extensions **104**. The accordion extensions **104** allow the actuator **100** to bend or flex when inflated, and help to define the shape of the actuator **100** when in an inflated state. The accordion extensions **104** include a series of ridges **106** and troughs **108**. The size of the accordion extensions **104** and the placement of the ridges **106** and troughs **108** can be varied to obtain different shapes or extension profiles.

(23) Although the exemplary actuator of FIGS. **1A-1D** is depicted in a “C” or oval shape when deployed, one of ordinary skill in the art will recognize that the present invention is not so limited. By changing the shape of the body of the actuator **100**, or the size, position, or configuration of the accordion extensions **104**, different sizes, shapes, and configurations may be achieved. Moreover, varying the amount of inflation fluid provided to the actuator **100** allows the retractor to take on one or more intermediate sizes or shapes between the un-inflated state and the inflated state. Thus, an individual actuator **100** can be scalable in size and shape by varying inflation amount, and an actuator can be further scalable in size and shape by replacing one actuator **100** with another actuator **100** having a different size, shape, or configuration.

(24) The actuator **100** extends from a proximal end **112** to a distal end **110**. The proximal end **112** connects to an interface **114**. The interface **114** allows the actuator **100** to be releasably coupled to a hub or other robotic components. The interface **114** may be made of a medically safe material, such as polyethylene, polypropylene, polycarbonate, polyetheretherketone, acrylonitrile-butadiene-styrene (“ABS”), or acetal homopolymer. The interface **114** may be releasably coupled to one or both of the actuator **100** and the flexible tubing **118**. The interface **114** may have a port for connecting to the actuator **100**. Different interfaces **114** may have different sizes, numbers, or configurations of actuator ports, in order to accommodate larger or smaller actuators, different numbers of actuators, or actuators in different configurations.

(25) The actuator **100** may be inflated with an inflation fluid supplied from an inflation device **120** through a fluidic connection such as flexible tubing **118**. The interface **114** may include or may be attached to a valve **116** for allowing fluid to enter the actuator **100** but preventing the fluid from exiting the actuator (unless the valve is opened). The flexible tubing **118** may also or Alternatively, attach to an inflator valve **124** at the inflation device **120** for regulating the supply of inflation fluid at the location of the inflation device **120**.

(26) The flexible tubing **118** may also include an actuator connection interface **122** for releasably connecting to the interface **114** at one end and the inflation device **120** at the other end. By separating the two parts of the actuator connection interface **122**, different inflation devices **120** may be connected to different interfaces **114** and/or actuators **100**.

(27) The inflation fluid may be, for example, air or saline. In the case of air, the inflation device

120 may include a hand-operated bulb or bellows for supplying ambient air. In the case of saline, the inflation device **120** may include a syringe or other appropriate fluid delivery system. Alternatively, or in addition, the inflation device **120** may include a compressor or pump for supplying the inflation fluid.

(28) The inflation device **120** may include a fluid supply **126** for supplying an inflation fluid. For example, the fluid supply **126** may be a reservoir for storing compressed air, liquefied or compressed carbon dioxide, liquefied or compressed nitrogen or saline, or may be a vent for supplying ambient air to the flexible tubing **118**.

(29) The inflation device **120** further includes a fluid delivery device **128**, such as a pump or compressor, for supplying inflation fluid from the fluid supply **126** to the actuator **100** through the flexible tubing **118**. The fluid delivery device **128** may be capable of supplying fluid to the actuator **100** or withdrawing the fluid from the actuator **100**. The fluid delivery device **128** may be powered by electricity. To supply the electricity, the inflation device **120** may include a power supply **130**, such as a battery or an interface to an electrical outlet.

(30) The power supply **130** may also supply power to a control device **132**. The control device **132** may allow a user to control the inflation or deflation of the actuator, e.g. through one or more actuation buttons **134** (or alternative devices, such as a switch). The control device **132** may include a controller **136** for sending a control signal to the fluid delivery device **128** to cause the fluid delivery device **128** to supply inflation fluid to, or withdraw inflation fluid from, the actuator **100**.

II. USER-ASSISTED ROBOTIC CONTROL SYSTEMS

(31) Exemplary embodiments encompass several aspects of robotic control systems. In particular, some embodiments relate to an overall architecture including a “human in the loop” form of supervisory control. Such a system may provide hardware and software to allow a user to interact with one or multiple robotic systems either locally or remotely, and may include, separately or in combination, improvements in robot control and in analysis systems for performing tasks such as data analytics and data mining, and other improvements.

(32) Further embodiments, which may be used separately or in conjunction with other embodiments described herein, relate to user interface design for robotic control systems. The exemplary user interfaces described herein may provide efficient and simple procedures for interacting with one or multiple robotic systems.

(33) Further embodiments, which may be used separately or in conjunction with other embodiments described herein, relate to improved visualization and three-dimensional locating and maneuvering, as well as camera and robot calibration, and other improvements.

(34) Still further, some embodiments (which may be used separately or in conjunction with other embodiments described herein) relate to modular robotic grippers that allow different types or configurations of grippers to be rapidly deployed in connection with a robotic system.

(35) The overall architecture is next described with reference to FIGS. 2A-2G. Unless otherwise noted, individual improvements described herein may be used separately or in combination with any other improvements.

A. Architecture Overview

(36) FIG. 2A is a block diagram depicting an overview of a robotic control architecture **200**. Although the exemplary architecture of FIG. 2A includes particular elements in particular arrangements or configurations, other elements, arrangements, and combinations may be employed without deviating from the scope of the present invention.

(37) According to exemplary embodiments, a user may interact with an electronic processing device **202** (e.g., a desktop computer, a laptop computer, a mobile phone, a tablet, a special-purpose kiosk, etc.). In some embodiments, the electronic device **202** may be provided with a touch screen or other form of haptic input for allowing the user to interact with an interface via touches and/or gestures. In other embodiments, the electronic device **202** may be provided with some other form

of input, such as a keyboard, a mouse, a video camera or instrumented glove for gesture-based control, a microphone and associated logic for performing voice control, etc. Interactions with the electronic device may be in the form of various control directives. A control directive may be any input from a user to the electronic device **202**, for example, and without limitation, a touch on a touch sensitive display with a finger or a stylus, the use of an input device to select a user interface element such as an icon or a menu item, an audio command, a motion-captured input, a motion of the electronic device, and so forth.

(38) Using the electronic device **202**, the user is able to interact with a robotic system or multiple different robotic systems. Preferably, the electronic device **202** allows the user to interact, via control directives, with multiple (e.g., three or more) robotic pick-and-place stations simultaneously. An exemplary electronic device for controlling robotic systems is depicted in FIG. 2B below, and further described in the next section relating to device interfaces.

(39) The electronic device **202** may include a wired or wireless network interface for connecting to a network **204**, such as the Internet. Through the network connection, the user may be capable of interacting with robotic systems in a remote location. For example, a user in their home or an office may control robotic systems in a remote warehouse, fulfillment center, or other location anywhere that is accessible to the network **204**.

(40) According to some embodiments, a human user may be supplemented by, or may be replaced by, artificially intelligent control logic. Accordingly, the system may be readily upgradeable as technology advances. For example, initially a user may fully control the grasp actions of the robotic system, including designating where the robotic gripper should deploy in order to grasp a target item.

(41) As visualization and object recognition technology improves, the system may recognize and/or differentiate between different target objects, allowing the user to simply tap or click on a target object to select it. The gripper may then automatically deploy in order to grasp the target. In such a scenario, a human user may still make decisions as to which targets to grasp, in which order, and where to release the target objects. As artificially intelligent controls improve further, machine logic may fully replace a human user in making these decisions.

(42) The network **204** may be the Internet, a public network, a private network, or any combination thereof. Examples of network architectures are described in more detail in connection with FIG. 8.

(43) Using the network, the electronic device **202** may interact with one or more local (e.g., local to the robotic system) processors **210-1**, **210-2** . . . **210-n**, where n is a positive integer. The local processors **210** may be in the form of any suitable computing device, such as a desktop computer, a server, a mini- or micro-computer, a laptop, a tablet, a mobile device, special-purpose hardware, etc. In some embodiments, each pick-and-place station **220** may be associated with one or more local processors **210** that are unique to the pick-and-place station (e.g., not shared among pick-and-place stations). In other embodiments, one or more local **210** processors may be associated with multiple pick-and-place stations, e.g. local processor **210-2** may be associated with pick and place stations **220-2** and **220-m**.

(44) The local processors **210** may receive incoming commands from the user's electronic device **202** and may relay appropriate control instructions or commands to the robotic system. Moreover, the local processors **210** may receive information from the robotic system (including from any sensors deployed in connection with the robotic system), and other components of the pick-and-place station such as the camera, the conveyor belts, the bins, etc. Information from the pick-and-place station **220** may be transmitted to the user's electronic device **202** to facilitate control of the robotic system.

(45) Furthermore, information from the local processor, the robotic systems, and/or the user device may be transmitted to a remote processor, depicted as a cloud backend **240** in FIG. 2A. The cloud backend **240** may be any suitable electronic device, such as a desktop computer, a server, a mini- or micro-computer, a laptop, a tablet, a mobile device, special-purpose hardware, etc. The cloud

backend **240** may collect data regarding deployed systems, and may use the data for a number of purposes.

(46) For example, the cloud backend **240** may collect snapshots from a camera **222** deployed in the pick-and-place station **220-1** (e.g., at predetermined intervals, or when predetermined events such as possible failure conditions, occur). The cloud backend **240** may further receive information from sensors deployed in conjunction with the pick station **220-1** and/or robotic sensors, such as from object range and depth sensors, grip quality detection and sensors, information about pneumatic/hydraulic flow rates and pressures, and so forth. The cloud backend **240** may also be provided with further information (automatically or manually), including failure rates of actuators, grippers, and other robotic systems, maintenance schedules and identified problems, software crashes, etc.

(47) The cloud backend **240** may perform data mining and analytics on the collected data. Based on the collected data, the cloud backend **240** may recommend changes to maintenance or replacement schedules, may suggest repairs or optimizations, and/or may automatically make (or may recommend) changes or improvements to the software control systems of the local processors **210** and/or the user's electronic device **202**, among other possibilities.

(48) For example, if the cloud backend **240** determines, based on one or more grasp detection sensors and/or range sensors, that the current robotic control software is not descending to an appropriate depth before attempting a grasp by closing a robotic gripper **234**, the cloud backend **240** may transmit an update or patch that changes the initial depth at which a robotic arm **232** deploys. Similarly, the cloud backend **240** may transmit changes to the gripper orientation, the motion path, the grasp force, etc.

(49) The update may be applied locally (e.g., to change the behavior of one robot) or globally (e.g., to change the behavior of a group of robots, a fleet of robots, or all robots accessible to the cloud backend **240**). For example, if 10 robots in a fleet of 10,000 attempted to grip a new object with varied success, then based on these teachings the entire fleet of 10,000 received an update that allowed all of the robots to successfully grasp the item.

(50) To these ends and others, the cloud backend **240** may further transmit over-the-air updates to the local processors **210** to upgrade or repair the control software for the robotic systems. The cloud backend **240** may also transmit over-the-air updates to the user's electronic device **202** for purposes of software upgrade or repair, among other possibilities.

(51) As noted above, the local processor(s) **210** may be associated with one or more pick-and-place stations **220** assigned to a particular user. For example, a given location, such as a warehouse or fulfillment center, may include many pick-and-place stations, with particular stations or groups of stations assigned to different users. As used herein, a “pick-and-place station”—or simply “pick station”—may represent many different applications where a robot arm and gripper manipulates objects, such as (but not limited to) bin-picking for warehouses, produce packaging, machining, inspections, etc. The following are but a few examples of applications for the technology described herein: Picking a casting of a metal part, for example a partially finished brass valve, from a pile and placing it in a fixture for machining Picking a melon from a pile and placing it in a peeler chunker machine. Picking a single piece of plastic of a particular polymer type (e.g. polyethylene, polypropylene, polyethylene terephthalate, etc.) out of a pile of mixed plastics at a recycling center and placing it in a bin of just that plastic type. Picking a piece of fruit from a tree and placing it in a crate.

(52) A simplified example of a pick station is depicted in FIGS. 2C and 2D.

(53) According to exemplary embodiments, each pick station, e.g. pick station **220-1**, may be associated with a robotic system, a source bin **226** in which target objects **229** are delivered to the pick station **220-1** to be grasped by the robotic system, a target bin **228** into which target objects **229** are delivered by the robotic system, and one or more cameras **222** (e.g., one or more cameras deployed above the source bin and/or the target bin). The source bin **226** and/or target bin **228** may

be delivered to the pick station by a person or by means of conveyor belts **224** or other suitable devices.

(54) One of ordinary skill in the art will understand that other configurations for a pick station are possible. For example, the target objects may not initially be deployed in a source bin, but could be delivered directly on the conveyor or in another manner that does not rely on conveyors. In one example, the target objects may be deployed on a mobile shelving system, where an entire shelf is moved to the pick station. Similarly, the target object may be delivered to a receptacle or other recipient location that does not include a target bin. The camera could be mounted to the side or below the container. Many other configurations are also possible within the scope of the present invention.

(55) In addition to a camera, other sensors may be present to detect objects in a bin or to detect objects as they enter or leave a bin. Such sensors may include but are not limited to weight sensors (load cell or force/torque sensor), accelerometers or other vibration sensors, three-dimensional depth cameras, photoelectric break beams, spectrometers, hyperspectral imagers, and laser range scanners.

(56) The robotic system may include a base **230**, to which a robotic arm **232** including one or more joints is attached. The base **230** and/or arm **232** may support pneumatic and/or hydraulic connections for actuating a soft robotic gripper **234**, and/or electrical and/or data connections for supporting sensors and other devices deployed in connection with the arm **232** or gripper **234**.

(57) The gripper **234** may include a hub for connecting the gripper **234** to the robotic arm **232**. One or more actuators (e.g., soft robotic actuators) may be attached to the hub to form the gripper. The hub may include connections, such as hydraulic or pneumatic connections, to deliver actuation fluid to the actuator(s), either separately or as a group.

(58) The hub may further include mounting points for sensors and other devices, and may include electrical and/or data connections for powering and communicating with the devices. For example, the hub may include a range sensor deployed in a location (e.g., the bottom of the hub, between the actuators) that allows the system to determine a range to a target object to be grasped. The hub may further include a camera to sense and visualize characteristics of a grasped object prior to, during, and after attempting a grasp of that object. The hub may further include an accelerometer or force detection sensor to determine sway characteristics, momentum, or inertia information relating to a grasped object (thereby allowing the system to adjust the manner in which the object is held or moved to improve holding or transport characteristics). Further embodiments may include one or more grasp or force sensors to determine how well or with how much force a gripper is holding a target object. The grasp or force sensors may be mounted on or deployed in the hub and/or actuators.

(59) In some embodiments, each actuator may be associated with a hub slidably attached to one or more rails that permit the actuators to move relative to each other. The rail may be attached to the robotic arm. The hubs may be manually adjusted along the rail, or may be automatically adjusted via a motor, pneumatics, hydraulics, etc. Accordingly, the spacing between the actuators and/or the configuration of actuators with respect to each other may be adjusted as needed. Exemplary grippers suitable for use with exemplary embodiments are further described in the section on modular grippers, below.

(60) In some embodiments, more than one hub may be connected to each other by rails, so as to support simultaneous grasping of multiple objects. The actuators associated with each hub may be controlled independently or as a single unit. The relative positions of the hubs may be adjusted along the rails, either manually or automatically via motor, pneumatics, hydraulics, etc.

(61) Target objects may be delivered to the pick station in a source bin. The source bin may be any receptacle suitable for conveying items to the pick station. An example of an interior view of a source bin including several target items is depicted in FIG. 2E. It is noted that, although particular examples are described in connection with a “bin,” this term may encompass many types of

receptacles, such as boxes, cases, etc.

(62) The source bin may be delivered to a location accessible to the robotic system, e.g. by a conveyor system. The robotic system may grasp one or more items in the source bin and deliver the items to a target location, such as a target bin that is also located in a position accessible to the robotic arm. The source bin may be moved away from the robotic system when empty and/or when the relevant items have been removed from the source bin, allowing the source bin to be replaced by a new source bin with new target items. Similarly, when the target bin is full or has received sufficient relevant items, the target bin may be moved away and/or replaced with a new target bin.

(63) One or more cameras or other visualization sensors may be mounted in a location in which the contents of the source bin and/or target bin may be viewed. For example, camera(s) may be mounted on a gantry above the source bin and/or target bin. Examples of camera systems are depicted in FIGS. 2F-2G.

B. Exemplary Interfaces

(64) The cameras and/or other sensors may be connected (wired or wirelessly) to the local processors **210**, which may relay a view of the camera (or other sensor data) to the cloud backend **240** and/or the user's electronic device **202**. An interface on the user device **202** may display the camera view, as shown in FIG. 3A.

(65) The interface **300** may include one or more icons or other interactive elements for configuring or controlling the robotic system. Interactive elements may be provided for other operations, for example, and without limitation, a configuration operation, a calibration operation, a grasping operation, a force-setting operation, a flutter operation, a rearranging operation, an actuator sequencing operation, a moving operation, an object selection operation, or a stirring operation. Selecting an interactive element via a control directive may cause a control command corresponding to the interactive element to be transmitted to the robotic system in a selected pick and place station. For some operations, further input may be needed before a control command is transmitted, in which case additional user interface elements and/or interactive elements may be presented to obtain the input.

(66) For example, the depicted example includes a configuration/calibration icon **302** in the bottom left corner that may correspond to a configuration or calibration operation. Interacting with the configuration/calibration icon **302** may cause one or more menu options to be displayed in the interface.

(67) One such menu item may be an option to perform calibration of the pick station, which may include calibrating the camera view in order to provide a canonical view of the source and/or target bin. Such a calibration procedure is described in the visualization section below. In some embodiments, the user may also calibrate the camera view by touching a location (such as the center of an object) on a touch-sensitive display and dragging the haptic contact to re-orient the display.

(68) The main display of the interface may show a view from one or more of the cameras at a particular pick-and-place station. For example, the camera may default to a camera over a source bin. As noted above, multiple pick stations may be accessible and controllable from a given user device. In order to switch between pick stations, the user may, for example, execute a swipe gesture on a touch-sensitive display to move between cameras and/or pick stations. In another example, the user may select an interactive element on the display that may be visible, as an icon or button element, or else hidden, as a region of the display such as one of the four edges of the display. Still further, the interface may display multiple pick station views on a single screen (e.g., in a tiled or cascaded view).

(69) Switching between views of different pick stations may also transfer control from one robotic system to another. For example, if the interface initially displays an image from a first pick station, user commands issued through the electronic device **202** may be issued to the robotic system at the first pick station. If the user then swipes to switch the view to a second pick station, commands

issued via the electronic device **202** may be issued to the robotic system at the second pick station. (70) In some embodiments, the interface may be capable of displaying views of the destination bins in addition to views of the source bins. This may allow, for example, for a user to designate a particular location to which a grasped object should be delivered. In another example, the user may designate an orientation for the target object. Alternatively, or in addition, the interface may display a view of a source bin, and allow the user to indicate which objects to grasp from the source bin. The system may then automatically deliver the grasp target to a predetermined target location (such as an expected location of a target bin).

(71) In either case, there may be multiple target locations accessible at a given pick station. Two or more icons or interactive elements may be presented on the interface to allow the user to choose between different target locations. For example, the embodiment depicted in FIG. 3A includes three icons **304-1**, **304-2**, and **304-3** (receptacles with downward facing arrows) to indicate different target locations. In this example, selecting one of the target location icons after selecting a gripper target may deliver the grasped object to a destination location associated with the selected icon.

(72) If the user is enabled to select a particular target location (e.g., via a camera view of a destination bin or receptacle), the user may indicate a target location through, for example, a tap or click at a location at which the target object is to be placed.

(73) Alternatively, or in addition, the system may allow a user to specify a location or a pattern that the robot should employ when delivering grasp targets in the future and/or a sequence or ordering for placing the objects in the designated positions. For example, the target receptacle may be an egg carton or apple carrier, in which objects are placed in a grid configuration. The user may select multiple points on the interface to designate locations in a grid into which object should be placed. The robotic system may first deliver an object to the first grid location, then the next target object may be delivered to a second grid location, etc. When all grid locations have been filled, the conveyor associated with the destination receptacle may move the filled receptacle out of the way and replace it with a new receptacle, and the process may be repeated.

(74) Optionally, the system may optimize or normalize the user's selections—for example, if the user selects points in a 4×4 grid, with each point separated by an average of three inches, the system may equalize the distance between the points so that they are evenly spaced. The system may also, or alternatively, auto-complete a grid or other pattern if the system recognizes that the user has entered more than a predetermined number of points corresponding to the grid or pattern.

(75) The user may interact with the camera view to indicate where and/or how a given object should be grasped. The user may indicate, through the interface, where the edges of a gripper target are located. The robotic system may then deploy the grasper to the targeted location and attempt to close the grasper around the indicated edges.

(76) Generally, the gripper target edges may be indicated in the interface using an exemplary gripper display element, as depicted in FIG. 3B.

(77) Referring to the top view in FIG. 3B, the interface element may include gripper target **328** visual elements that represent the ends of the actuators in the gripper. A location of the gripper targets **328** may correspond to a location where the gripper will position the actuators at the beginning of a grasping operation of a physical target object. The interface element may include a line **330** representing an axis of the gripper (e.g., an axis along which the actuators are arranged).

(78) The interface element may also include one or more lines or tick marks along the axis line representing a minimum or maximum extent **322** to which the grippers can deploy (e.g., the minimum or maximum spacing accessible to the actuators, which may include spacing accessible by negative inflation of the actuators). A center point **324** of the axis may be marked between the marks indicating the minimum or maximum extent(s) of the actuators. The center point may be indicated by a point, a tick mark, a dashed line, etc.

(79) In some embodiments, the minimum/maximum extent may be the minimum/maximum extent accessible with the grippers in their current configuration; for example, if the grippers are mounted

on rails, the minimum/maximum extent may reflect the minimum/maximum distance to which the grippers can be extended given their current spacing along the rails. In other embodiments, the minimum/maximum extent may be the minimum/maximum extent accessible by the grippers if the grippers are reconfigured (e.g., the minimum/maximum extent accessible along the rails, even if the actuators are not currently deployed at their minimum/maximum spacing). The former embodiment may be useful in situations in which the actuators are manually adjustable along the rails, potentially requiring that a user interact with the robotic system to physically move the actuators. The latter embodiment may be well suited to applications where the actuators can be automatically moved along the rails by means, e.g., of a motor, pneumatic, or hydraulic system.

(80) In still further embodiments, the system may indicate both the minimum/maximum extent that is currently accessible and the minimum/maximum extent that would be possible with reconfiguration. For example, the system may mark the currently accessible minimum/maximum extent with a solid line and/or a line in a first color or shade, and may mark the minimum/maximum possible extent with a further dashed line **326** and/or a line in a second color or shade. The further dashed or shaded line **326** may extend along the axis and may terminate in two further marks indicating the minimum/maximum possible extent.

(81) According to exemplary embodiments, a user may interact with a haptic display through a one-finger or two-finger gesture to indicate where the robotic system should deploy the actuators of the gripper. The user gesture may further include more than two fingers to indicate deployment of the actuators, especially in embodiments that include more than two actuators. The interface may display a number of gripper targets **328** corresponding to the location at which the actuators (in some embodiments, the location at which the distal tips of the actuators) will be deployed. The interface may also include an indication **330** of the spacing between the gripper targets (e.g., a quantitative indication).

(82) FIG. **3C** depicts an exemplary gripper display element imposed on a camera view to enable a user to set the gripper targets, for example, for a grasping operation.

(83) In a one-finger embodiment, a user may make initial haptic contact with a touch screen (or may otherwise indicate an initial contact point, e.g. using a mouse or other input option). In one example, this point may serve as the initial gripper target. The user may then drag the initial input to a second location, which may serve as a second gripper target. The indicator on the interface may reorient and/or reconfigure itself as the user moves the initial contact. Alternatively, the user's initial contact may identify a center point at which the user intends to deploy the gripper (e.g., corresponding to the center mark identified above). When the user drags the initial contact, the amount of distance dragged may identify a spacing between the gripper targets.

(84) In a two-finger embodiment, a user's initial haptic contact with the interface may identify the location of the first gripper target, whereas a second haptic contact may identify the location of the second gripper target. In another embodiment, the first contact may identify the center location and the second contact may identify the location of one of the gripper targets—the other gripper target location may be inferred based on the distance between the center point and the explicitly specified gripper target.

(85) In either case, the system may optionally displace the displayed gripper targets by a predetermined distance away from the location of the haptic contact, if the contact is specified through a touch display. This may allow the user to view the gripper targets on the display without the gripper targets being obstructed by the user's hand or fingers. In further embodiments, the system may allow a user to zoom in on the interface in order to provide for more fine-grained selection of gripper targets.

(86) In some embodiments, the interface may enforce a limit such that the user is not permitted to move the gripper targets beyond the indicated maximum extents. For example, once the user moves the gripper targets to the indicated maximum extents, further movement beyond the indicated maximums may not result in further movement of the gripper targets (e.g., the gripper targets may

“stick” to the maximum extents, as shown in FIG. 3D below).

(87) In other embodiments, the interface may allow a user to set gripper targets that extend beyond the maximum extent indicated on the interface element. Due to the deformable nature of soft robotic grippers, the soft actuators may be capable of making sufficient contact with an object (using a grasp surface of the actuator) such that the gripper is capable of grasping an object larger in size than the maximum extent. Accordingly, the system may attempt to grasp the target object as closely as possible at the gripper target points indicated by the user; if a suitably strong grip cannot be achieved after a predetermined number of attempts, then the system may abort the attempt. Optionally, the system may attempt to stir the bin and/or otherwise rearrange the contents of the bin in an attempt to present a more graspable face of the target object to the user; such techniques are described in more detail below.

(88) When a user attempts to set gripper targets that extend beyond the maximum extent, the interface may indicate that the user is attempting to do so and may indicate the amount of extension beyond the maximum extent that is being requested. The interface may, for example, set the gripper targets beyond the lines for the maximum extent and indicate the amount of the additional extension requested, e.g., by dashed lines or lines of a different color, as indicated by the bottom drawing in FIG. 3B and in FIG. 3E.

(89) An alternative interface element is depicted in FIGS. 3F-3G. In these examples, the circles 334 represent locations on the haptic interface that a user has touched (i.e., haptic contacts), whereas the rounded bars 332 indicate the grasp capabilities for the tool. Depending on whether the user is attempting to grip below the minimum grasp capabilities of the tool (FIG. 3F) or beyond the maximum grasp capabilities of the tool (FIG. 3G), the rounded bars may move to the respective minimum/maximum locations to show where the actuators will actually grasp the object. When an attempted grasp is in-bounds (FIG. 3H), the rounded bars may fall directly on top of the haptic contact points to indicate that the gripper will deploy the actuators at the requested locations.

(90) Optionally, after initially setting the gripper targets, the user may adjust the gripper targets by interacting with the interface (e.g., selecting one or more gripper targets and moving them, moving the center point, etc.).

(91) The camera view may be a two-dimensional view, and accordingly the user may be capable of designating the initial placement of the actuators in two dimensions (e.g., the x/y dimensions). However, the robotic system may operate in three-dimensional space. The robotic system may move the actuators to the designated x/y location along a path in the z-direction (as described in more detail in the visualization section, below) until the gripper is in close proximity to the target object. For example, the gripper may be provided with a range sensor that may detect a distance to a target to be grasped. When the sensor indicates that the object is in the vicinity of the gripper, the gripper may be actuated to grip the object.

(92) In some embodiments, the center point of the gripper may be moved towards the x/y location of the center point indicated on the display. The actuators may initially be unactuated. Alternatively, if the gripper spread indicated on the interface would require negative actuation/inflation in order to be achieved, the actuators may be fully negatively inflated or partially negatively inflated by an appropriate amount.

(93) In other embodiments, the grippers may be partially actuated in order to approach the designated x/y locations with the grippers deployed as closely as possible to the locations indicated on the interface. These embodiments may be useful in situations where the source bin is relatively cluttered or full, and may allow for the gripper to approach the target object as precisely as possible while avoiding neighboring objects.

(94) In some embodiments, the user may be provided with an interactive element on the interface to activate a “flutter” effect on the actuators. When the actuators are fluttered, they may rapidly partially actuate and de-actuate while approaching the grasp target, thereby allowing nearby objects to be moved out of the way as they approach the grasp target. In some embodiments, the system

may automatically flutter the actuators, either as a matter of course or under certain conditions (e.g., when a cluttered environment is detected).

(95) Furthermore, the user may be permitted to flutter the actuators while the actuators grasp a target object (e.g., while the actuators are in contact with the target, to slightly manipulate the object so that more of the gripping surface is exposed). This motion may be followed by a gripping action to acquire the object for transport out of the bin.

(96) The above-described concept may be modified to suit other gripper types. For example, if a three- or four-actuator gripper is employed, the system may display an appropriate number of gripper targets in an appropriate configuration. Similar interactions to those described above may be used to set the gripper targets on the interface. For example, the user may initially place a haptic contact at a center location, and then move their initial contact or place an additional contact to move the gripper targets out from the center or in towards the center. Alternatively, or in addition, the user may tap at various locations on the screen to indicate preferred locations for the gripper targets. The system may determine a deployment configuration for the actuators that is compatible with the physical constraints of the gripper but also matches the preferred locations as closely as possible.

(97) In some embodiments, the interface may also include an interactive element allowing a user to designate a force to be applied to the grasp target. For example, the system may pop up a force menu when the gripper targets are set. Alternatively, or in addition, the system may register a three-dimensional touch force on a haptic display, or may include an input field allowing the grasp force to be specified. In another example, a drop down menu may allow a user to set a grasp force to be applied until changed.

(98) The grasp force may be specified quantitatively (e.g., 10N) and/or qualitatively (e.g., low, medium, or high). In some embodiments, the grasp force may be selected from among a predetermined set of possible grasp forces. In other embodiments, the grasp force may be specified within a predetermined range between the minimum grasp force and the maximum grasp force the gripper is capable of (or configured for) exerting.

(99) The interface may also permit the user to queue different grasps. For example, upon specifying the gripper targets and/or selecting a destination icon to specify a destination for the target object (or otherwise finalizing the grasp), the robotic system may begin executing the grasp. It may take some time for the robot to grasp the target object and move it to its destination. During the time that the robot is otherwise occupied, the user may interact with the interface in the same manner as discussed above to indicate further gripper targets. After the robotic system delivers the initial target object to the destination bin, the robotic system may continue to execute further grasps in the order specified. Queues may be specified in other ways, as well.

(100) When the robotic system is executing a grasp, the system may obscure the view of the source or destination bin due to the fact that the robotic arm may be present between the camera and the bin. As a result, the user may not be able to see the camera view in order to queue new gripper targets. This problem may also arise in other situations, such as when the robotic system is executing a calibration routine. In order to address this problem, the system may be configured to capture an image of the source or destination bins (e.g., at predetermined intervals, or just before the robot executes a pick) while the robotic arm is out of view of the camera. This image may then be used in the interface (either temporarily serving as the camera view, or serving as a ghost image partially superimposed on the current camera view while the robotic arm obscures the view) to allow the user to continue to specify gripper targets while the robotic arm is in the way. The “ghosted” image also aids the user in identifying objects that moved, e.g., due to the previous pick action of the gripper, because such objects may become partially transparent like the robotic arm. Objects that remain undisturbed and do not move may appear as normal, opaque objects in the user interface.

(101) In some embodiments, the system may allow a user to specify actuator sequencing; this may

be done before, during, or after the gripper targets are specified. For example, the order in which a two-finger gesture is executed may be the order in which the grippers are deployed. Alternatively, the user may specify gripper targets and then select one of the gripper targets to specify the order in which the corresponding actuator should be deployed. When the robotic system approaches the grasp target, the actuators may be actuated independently in the specified order.

(102) Further embodiments may allow for additional gestures to perform specified action. For example, a flicking gesture or some other indication, may instruct the robotic system to rearrange an object or nudge the object out of the way. In another example, a user may draw a circle or execute another gesture on the interface, which may instruct the robotic system to stir the contents of the bin. Accordingly, target objects can be rearranged in order to expose buried objects, to reorient target objects to expose a more graspable surface, and/or to move a target object away from a group of objects, a wall, or a corner of the source bin.

(103) In some embodiments, the user may define the path that the robotic arm takes in order to approach the target object. For example, the user may interact with the robotic system using a glove embedded with sensors, or a camera capable of three-dimensional motion tracking. The user may move their own arm along the path desired for the robotic arm. The sensor(s)/camera may analyze the path of the user's arm and transmit suitable instructions to the robotic arm defining a corresponding path that the robotic arm should take on the way to the grasp target. A similar technique may be used to define the deployment order of actuators, gripper force, etc.

(104) If visualization systems and/or intelligence permit, the interface may allow the user to select a recognized object using a tap or sustained contact. The user may drag the object to a designated target (e.g., a target icon corresponding to a desired destination bin).

(105) Further improvements in visualization systems are described in the next section.

C. Vision Systems

1. Calibration

(106) Exemplary embodiments provide techniques for quickly calibrating either or both of the robotic system and the visualization system, including the cameras.

(107) Typically, the source and destination bins are delivered to an area (e.g., on top of a conveyor belt) whose base is at a relatively fixed location in the horizontal direction (e.g., along the z-plane of an x-y-z coordinate system). This may simplify grasping tasks, because the robot can assume that target objects are likely somewhere between the bin base and the top of the bin. The robotic arm may then be dropped a certain distance before control is turned over to logic based on a range finder in the gripper that performs more fine control. Moreover, the source and/or destination bins may be delivered to approximately the same area each time; accordingly, the robotic system may have a general awareness (e.g., within a certain range) of where in the x/y plane the bins may be located.

(108) However, the robotic arm may occasionally fall out of calibration. In order to recalibrate the robotic arm, a calibration tool may be mounted to the gripper (e.g., a pointed rod for touching relatively precise locations; see the modular gripper example below). The robotic system may be configured to make contact with predetermined calibration elements in order to reset the robotic arm's understanding of the three-dimensional space in which the robotic arm operates, e.g. in an x/y/z or Cartesian reference frame. The robotic arm may further be recalibrated by moving into a predetermined home position and/or using various forms of range detection (e.g., a haptic contact sensor at the end of the calibration tool, a range sensor, etc.). Various sensors deployed in connection with the robotic system (e.g., the range sensor) may also be recalibrated during this process.

(109) In order to perform recalibration, the bins deployed to the pick-and-place station **220** may be provided with a calibration sheet **402** in a predetermined location, for example, at the bottom of a bin **404** as shown in FIG. **4A**.

(110) Instead of being blank, a calibration sheet **410** may include a predetermined pattern of visual

elements, as shown, for example, in FIG. 4B. The robotic system may be configured to use the camera image and/or one or more sensors to cause the calibration tool to contact the visual elements in sequence in order to recalibrate the x-, y-, and/or z-positioning of the robotic system. In the illustrated example, the robotic system may move the calibration tool to points P.sub.1, P.sub.2, P.sub.3, and P.sub.4 in sequence.

(111) The system may perform a calibration check at a variety of intervals using a target located at a common way point/safe height position for the robot. Accordingly, the range sensor may be checked against a target at a known distance (single height check) from a robot position (e.g., at the start of every pick cycle, or after a set interval of pick cycles, or after a difficult pick attempt, such as an attempt where gripping retries were executed) to verify that the range sensor is reading within a prescribed tolerance band. Similarly, a periodic single height range sensor check may be used to verify there is no foreign object present, e.g. debris blocking or affecting the range sensor accuracy.

(112) If an anomaly is observed in the range sensor reading, the robot may move to a gripper cleaning position at a station within the work envelope of the robot that provides an external gripper cleaning. For example, the station may provide a blow off function (e.g. an air puff plus vacuum) and a target-to-range sense to confirm the sensor protective cover is cleaned off. Failure to fix the issue by cleaning the sensor protective cover may indicate some type of sensor malfunction or drift, allowing a repair or maintenance technician to catch the issue before failed attempts are made at picking objects. The cleaning position may also, or alternatively, be the position above which the robot parks before or after a pick cycle, where the range sensor check is repeatedly performed.

(113) According to some embodiments, the calibration patterns may also, or alternatively, be used to calibrate or canonicalize the camera(s) or visualization systems. For example, the bins may not necessarily be delivered to the pick station perfectly oriented—new bins may be skewed by various amounts each time they are delivered to the station.

(114) Because it may be difficult for a user to interact with a system in which the view of the bins appears differently each time, the camera may use the calibration pattern and/or other cues (e.g., the edges of the bins as detected by edge detection software) in order to manipulate the view (e.g., skewing or reorienting portions or all of the image) in order to offer a standardized or canonical perspective. The system may canonicalize the view on a regular basis, upon the occurrence of predetermined events (e.g., when a new bin is delivered to the pick-and-place station, or after each object picking by the gripper), and/or based on a command from the user.

2. 3D Locating

(115) Robotic systems move in a three-dimensional space, so detecting that movement relative to grasp targets and place locations may be important. Some embodiments may include a depth camera capable of measuring the three-dimensional location and shape of objects in the camera field of view. Other embodiments, such as the exemplary embodiment described below, may instead use a two-dimensional camera with improvements to support three-dimensional locating and robotic control while still obtaining acceptable three-dimensional grasp results.

(116) When a two-dimensional camera provides a top-down view from a central location, the camera may provide an estimate of an x/y location at which an object is located but may not be able to localize the object in the z-direction. An example of a top-down image captured by a centralized camera is shown in FIG. 4C, which shows a bin **404** and a top-down view of an object **420**.

(117) The illustrated view of the object **420** may be the result of a relatively tall object located close to the camera, or a relatively short object located further away from the camera, as shown in a side view in FIG. 4D.

(118) In this example, a vector **442** drawn from the camera lens **422** to the top of the object **420** passes through both the relatively close location **444** (having a relatively large height in the z-plane) and the relatively far location **446** (having a relatively small height in the z-plane). With a

two-dimensional camera system, it may not be possible to determine from the image alone whether the object **420** is relatively tall and located close to the camera (**420-1**) or is relatively short and located far from the camera (**420-2**).

(119) Accordingly, if the system attempts to make a guess as to the precise x/y location of the target object and the robotic arm approaches the object from directly above the guessed location, it is possible that the robotic arm will drop towards the object but have nothing below the gripper to grasp (as demonstrated by the gripper representations **434-1** directly above the target object **420-1**, and gripper **434-2** above target object **420-2** in FIG. 4D).

(120) In order to avoid this problem, the robotic arm may instead approach the object along the vector **442** drawn from the camera **422** to the target object (as shown by the angled gripper representation **448** in FIG. 4D). This may be accomplished, for example, based on the x/y position of the gripper targets **328** and/or center point **324** as designated in the interface, and a known location of the camera relative to the gripper targets and/or center point.

(121) This approach ensures that the gripper will eventually approach the target object, regardless of the target object's height. A range detector in the robotic arm may detect a range to the target object, and actuate the actuators when it is detected that the target object is in range of at least the tips of the actuators.

(122) In some cases, it may be difficult to obtain an accurate range based on the range sensor. For example, many range sensors work by transmitting electromagnetic waves toward a target object and receiving the reflected waves back at a sensor. However, some objects scatter electromagnetic waves, making it difficult to achieve an accurate range reading.

(123) Accordingly, the robotic system may lower the grasper until the range sensor indicates that the grasp target is within reach. This may represent a first predetermined drop/approach distance. The system may then attempt to grasp the object by actuating the actuators. If one or more sensors (e.g., contact sensors such as a contact-sensitive skin in the actuators, force detection sensors, the camera, etc.) detect that the object has not been sufficiently grasped (e.g., the object is not held by the actuators, or the object is not held with sufficient force), the robotic system may descend by a predetermined amount from the first predetermined drop distance and try again. This process may be repeated until a sufficient grasp is achieved, or until the attempt is aborted (e.g., after a predetermined number of failed attempts). The predetermined drop distance and predetermined amount of descent may be automatically or manually reconfigured.

(124) In some embodiments, other parameters may also, or alternatively, be adjusted. For example, the system may adapt the force applied to the target object based on the grasp quality (when grasp quality is detected to be poor, the system may apply increased force by supplying further inflation fluid to the actuators, and vice versa).

(125) Typically, the system will attempt to grasp the object as specified by the user in the interface. For example, the system may rotate the gripper base to deploy the actuators at the angle and/or distance as specified for the gripper targets in the interface. However, this behavior may be overridden in certain conditions.

(126) For example, the system may adapt the grasp location based on grasp quality. If the sensors and/or camera detect a poor grip, the gripper may be rotated and/or translated in an attempt to grab a longer or otherwise more graspable surface of the target object. In another example, the system may alter the rotation of the gripper and/or grasp location to avoid collision with the bin or neighboring objects. The system may adapt the gripper differently depending on the particular type of collision to be avoided. For example, when approaching the side of a bin, the gripper may reorient itself so as to deploy the axis of the actuators either parallel or at a ninety-degree angle to the side of the bin. This determination may be made depending on the configuration of the object to be retrieved. On the other hand, when approaching the corner of the bin, the system may orient the gripper so that one actuator approaches the corner while another is disposed towards the interior of the bin. Other approaches and/or configurations are also possible.

(127) In some cases, the pick may be aborted altogether if the system determines that a collision with the side of the bin is inevitable and/or if the user requests that the robotic arm deploy the gripper to a location outside the walls of the bin or to some other off-limits location.

(128) In some embodiments, the robotic gripper may be mounted on a telescoping or fixed rod to reduce the diameter of the robotic arm in the vicinity of the rod. The system may manipulate the path at which the gripper approaches the target object so that the reduced-diameter portion is in the vicinity of the bin edge at given points along the path. In this way, the gripper may be able to approach target objects in relatively confined environments that may not be possible without the use of the rod.

3. Vision Hardware

(129) Further embodiments provide improvements in vision hardware for robotic systems. Many range finding sensors employ a transmitter/receiver pair, in which a signal (e.g., an electromagnetic signal, such as a radio wave or beam of infrared light) is transmitted, bounced off an object, and then received.

(130) For example, a range sensor may be mounted in a hub of a robotic gripper, as shown in FIG. 5A (where the actuators **502** are present in at the left and right, the rectangular bottom portion represents the hub **504**, and the dotted lines **506** represent electromagnetic energy transmitted from the transmitter **508a** and received at the receiver **508b**).

(131) When exposed to real-world use, the range sensor may become obscured by dirt or damaged by objects in the environment. Accordingly, in order to protect the range sensor and/or facilitate cleaning, the range sensor may be covered by a layer **510** of transmissive material, such as glass or plastic.

(132) However, a problem may arise in that the covering layer **510** may not be fully transparent to the electromagnetic energy transmitted by the transmitter. Accordingly, the energy from the transmitter may be reflected back by the covering layer and may be received at the receiver, resulting in erroneous range readings, as shown by line **507** in FIG. 5B.

(133) In order to alleviate this problem, a barrier **512** may be placed between the transmitter and receiver. The barrier **512** may be made of any suitable material that will block the energy from the transmitter and prevent the energy from being received at the receiver. The scale in FIG. 5B is exaggerated for illustration purposes.

(134) The barrier may be, for example, of a one-piece or two-piece design for use on emitter/receiver type range sensors. In either case, the designs primarily provide a separation barrier between the emitter and receiver objectives or apertures on the sensor. In some embodiments, the barrier may span the entire gap between the sensor outer surface and the inner surface of the protective cover; in other embodiments, the barrier may span part of the gap.

(135) In the case of a two-piece design, a barrier shape that surrounds the outer perimeter of the sensor may be attached around the sensor, and a matching barrier shape with the addition of a dividing rib or wall aligned to fit between the emitter and receiver of the range sensor may be adhered to the protective cover. When the protective cover is then installed with the two barrier elements in place, the range sensor effectively becomes boxed in and protected from scattered electromagnetic radiation, and has a barrier in place to separate and isolate the emitter and receiver objectives from each other.

(136) An example of a barrier suitable for use with exemplary embodiments is depicted in FIGS. 5C and 5D. As shown in FIG. 5C, the barrier may include two parts below the covering, one part **520** provided to fit the around the outer edges of the sensor and to support a second part **522**, which provides the barrier in a precise location relative to the sensor transmitter and receiver. FIG. 5C shows the two parts in isolation, whereas FIG. 5D shows one part deployed on a robotic hub and surrounding a range sensor **524**. The other part **522** mates with the first part **520** and positions the barrier appropriately.

(137) Using the exemplary barrier, the range detector may be protected from environmental factors

while nonetheless exhibiting increased detection accuracy.

D. Gripper Configuration

(138) Still further embodiments may include a modular gripper that is capable of being quickly swapped out (either entirely, or specific parts) for another suitable modular gripper. This may allow, for example, for grippers to be swapped for others with different configurations of electronics and/or actuators.

(139) FIGS. **6A-6D** show different perspective views of an assembled modular gripper **600**. As shown in the front perspective view of FIG. **6A**, the gripper **600** may include one or more ports **602** for supplying and/or removing inflation fluid to the actuators and for supplying electrical and/or data connections to sensors or processors. In some embodiments, different ports may feed different actuators such that the actuators are actuatable independently. One or more pneumatic or hydraulic lines may be attached to the ports in order to supply the inflation fluid to the gripper

(140) Note also, in the bottom view of FIG. **6D**, that the modular gripper **600** may include mounting points **604** for sensors, such as range sensors.

(141) The modular gripper **600** may be disassembled so that one or all pieces may be swapped, repaired, or replaced. FIG. **6E** shows the modular gripper **600** of FIG. **6A** when disassembled. The rod of FIG. **6E** is a calibration tool **610**, as mentioned in previous sections, which may be mounted to a hub base part **622** in place of the actuator assembly **624**.

(142) The pieces depicted may be secured by a combination of friction and/or fastening mechanisms, such as screws. To assemble the pieces, the hub base **622** may be secured to the robotic arm via a mount **626**, as depicted in FIG. **6L**; see also FIG. **6J** (side view) and **6K** (internal view). A front support **628** and a rear support **630** may be slid into corresponding slots in the hub base **622**. The actuator base **632** may be attached to the bottom of the assembly, e.g. to one or more of the hub base **622**, front support **628**, or the rear support **630**, through screws or a quick-release fitting.

(143) According to some embodiments, the hub base **622** may be configured to remain attached to a wrist flange of the robotic arm when other pieces of the modular gripper is/are removed, swapped, or reconfigured. Therefore, the gripper can be easily reconfigured without affecting the alignment of the gripper to the robotic arm.

(144) Similarly, the hub base **622** may also provide an attachment provision for the robot calibration rod **620**, which can be installed and removed without impacting the alignment of the gripper to the robot. This is due to the fact that the gripper is removed from the mounting structure while the calibration rod is installed, and then the gripper is re-installed into specially-configured grooves with alignment and locking features (or other elements) that allow the gripper to be accurately and repeatedly installed.

(145) The modular design of the gripper allows various combinations of side panels (allowing for different controls connections) and even overall gripper designs to be installed on the same mounting structure, and also allows for grippers with different sensing, internal electronics, and even different actuator arrangements to be installed while preserving the same alignment to the robot wrist.

(146) Any or all of the components may be swapped with components having a different configuration. For example, the actuator base may be swapped for a different actuator base having a different type of sensor, while still maintaining the rest of the modular gripper components. In another example, a new actuator base having a different number and/or arrangement of actuators may be swapped in, and a different front support having different inlet and/or outlet ports may also be swapped in to provide for independent actuation of the actuators.

III. COMPUTING SYSTEM AND NETWORK IMPLEMENTATION

(147) The above-described techniques may be embodied as instructions on a non-transitory computer readable medium or as part of a computing architecture. FIG. **7** illustrates an embodiment of an exemplary computing architecture **700** suitable for implementing various embodiments as

previously described. In one embodiment, the computing architecture **700** may comprise or be implemented as part of an electronic device, such as a computer **701**. The embodiments are not limited in this context.

(148) As used in this application, the terms “system” and “component” are intended to refer to a computer-related entity, either hardware, a combination of hardware and software, software, or software in execution, examples of which are provided by the exemplary computing architecture **700**. For example, a component can be, but is not limited to being, a process running on a processor, a processor, a hard disk drive, multiple storage drives (of optical and/or magnetic storage medium), an object, an executable, a thread of execution, a program, and/or a computer. By way of illustration, both an application running on a server and the server can be a component. One or more components can reside within a process and/or thread of execution, and a component can be localized on one computer and/or distributed between two or more computers. Further, components may be communicatively coupled to each other by various types of communications media to coordinate operations. The coordination may involve the uni-directional or bi-directional exchange of information. For instance, the components may communicate information in the form of signals communicated over the communications media. The information can be implemented as signals allocated to various signal lines. In such allocations, each message is a signal. Further embodiments, however, may alternatively, employ data messages. Such data messages may be sent across various connections. Exemplary connections include parallel interfaces, serial interfaces, and bus interfaces.

(149) The computing architecture **700** includes various common computing elements, such as one or more processors, multi-core processors, co-processors, memory units, chipsets, controllers, peripherals, interfaces, oscillators, timing devices, video cards, audio cards, multimedia input/output (I/O) components, power supplies, and so forth. The embodiments, however, are not limited to implementation by the computing architecture **700**.

(150) As shown in FIG. 7, the computing architecture **700** comprises a processing unit **702**, a system memory **704** and a system bus **706**. The processing unit **702** can be any of various commercially available processors, including without limitation an AMD® Athlon®, Duron® and Opteron® processors; ARM® application, embedded and secure processors; IBM® and Motorola® DragonBall® and PowerPC® processors; IBM and Sony® Cell processors; Intel® Celeron®, Core (2) Duo®, Itanium®, Pentium®, Xeon®, and XScale® processors; and similar processors. Dual microprocessors, multi-core processors, and other multi-processor architectures may also be employed as the processing unit **702**.

(151) The system bus **706** provides an interface for system components including, but not limited to, the system memory **704** to the processing unit **702**. The system bus **706** can be any of several types of bus structure that may further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local bus using any of a variety of commercially available bus architectures. Interface adapters may connect to the system bus **706** via a slot architecture. Example slot architectures may include without limitation Accelerated Graphics Port (AGP), Card Bus, (Extended) Industry Standard Architecture ((E)ISA), Micro Channel Architecture (MCA), NuBus, Peripheral Component Interconnect (Extended) (PCI(X)), PCI Express, Personal Computer Memory Card International Association (PCMCIA), and the like.

(152) The computing architecture **700** may comprise or implement various articles of manufacture. An article of manufacture may comprise a computer-readable storage medium to store logic. Examples of a computer-readable storage medium may include any tangible media capable of storing electronic data, including volatile memory or non-volatile memory, removable or non-removable memory, erasable or non-erasable memory, writeable or re-writable memory, and so forth. Examples of logic may include executable computer program instructions implemented using any suitable type of code, such as source code, compiled code, interpreted code, executable code, static code, dynamic code, object-oriented code, visual code, and the like. Embodiments may also

be at least partly implemented as instructions contained in or on a non-transitory computer-readable medium, which may be read and executed by one or more processors to enable performance of the operations described herein.

(153) The system memory **704** may include various types of computer-readable storage media in the form of one or more higher speed memory units, such as read-only memory (ROM), random-access memory (RAM), dynamic RAM (DRAM), Double-Data-Rate DRAM (DDRAM), synchronous DRAM (SDRAM), static RAM (SRAM), programmable ROM (PROM), erasable programmable ROM (EPROM), electrically erasable programmable ROM (EEPROM), flash memory, polymer memory such as ferroelectric polymer memory, ovonic memory, phase change or ferroelectric memory, silicon-oxide-nitride-oxide-silicon (SONOS) memory, magnetic or optical cards, an array of devices such as Redundant Array of Independent Disks (RAID) drives, solid state memory devices (e.g., USB memory, solid state drives (SSD) and any other type of storage media suitable for storing information. In the illustrated embodiment shown in FIG. 7, the system memory **704** can include non-volatile memory **708** and/or volatile memory **710**. A basic input/output system (BIOS) can be stored in the non-volatile memory **708**.

(154) The computing architecture **700** may include various types of computer-readable storage media in the form of one or more lower speed memory units, including an internal (or external) hard disk drive (HDD) **712**, a magnetic floppy disk drive (FDD) **714** to read from or write to a removable magnetic disk **716**, and an optical disk drive **718** to read from or write to a removable optical disk **720** (e.g., a CD-ROM or DVD). The HDD **712**, FDD **714** and optical disk drive **720** can be connected to the system bus **706** by an HDD interface **722**, an FDD interface **724** and an optical drive interface **726**, respectively. The HDD interface **722** for external drive implementations can include at least one or both of Universal Serial Bus (USB) and IEEE 694 interface technologies.

(155) The drives and associated computer-readable media provide volatile and/or nonvolatile storage of data, data structures, computer-executable instructions, and so forth. For example, a number of program modules can be stored in the drives and memory units **708**, **712**, including an operating system **728**, one or more application programs **730**, other program modules **732**, and program data **734**. In one embodiment, the one or more application programs **730**, other program modules **732**, and program data **734** can include, for example, the various applications and/or components of the communication system **500**.

(156) A user can enter commands and information into the computer **701** through one or more wire/wireless input devices, for example, a keyboard **736** and a pointing device, such as a mouse **738**. Other input devices may include microphones, infra-red (IR) remote controls, radio-frequency (RF) remote controls, game pads, stylus pens, card readers, dongles, finger print readers, gloves, graphics tablets, joysticks, keyboards, retina readers, touch screens (e.g., capacitive, resistive, etc.), trackballs, trackpads, sensors, styluses, and the like. These and other input devices are often connected to the processing unit **702** through an input device interface **740** that is coupled to the system bus **706**, but can be connected by other interfaces such as a parallel port, IEEE 694 serial port, a game port, a USB port, an IR interface, and so forth.

(157) A monitor **742** or other type of display device is also connected to the system bus **706** via an interface, such as a video adaptor **744**. The monitor **742** may be internal or external to the computer **701**. In addition to the monitor **742**, a computer typically includes other peripheral output devices, such as speakers, printers, and so forth.

(158) The computer **701** may operate in a networked environment using logical connections via wire and/or wireless communications to one or more remote computers, such as a remote computer **744**. The remote computer **744** can be a workstation, a server computer, a router, a personal computer, portable computer, microprocessor-based entertainment appliance, a peer device or other common network node, and typically includes many or all of the elements described relative to the computer **701**, although, for purposes of brevity, only a memory/storage device **746** is illustrated.

The logical connections depicted include wire/wireless connectivity to a local area network (LAN) **748** and/or larger networks, for example, a wide area network (WAN) **750**. Such LAN and WAN networking environments are commonplace in offices and companies, and facilitate enterprise-wide computer networks, such as intranets, all of which may connect to a global communications network, for example, the Internet.

(159) When used in a LAN networking environment, the computer **701** is connected to the LAN **748** through a wire and/or wireless communication network interface or adaptor **752**. The adaptor **752** can facilitate wire and/or wireless communications to the LAN **748**, which may also include a wireless access point disposed thereon for communicating with the wireless functionality of the adaptor **752**.

(160) When used in a WAN networking environment, the computer **701** can include a modem **754**, or is connected to a communications server on the WAN **750**, or has other means for establishing communications over the WAN **750**, such as by way of the Internet. The modem **754**, which can be internal or external and a wire and/or wireless device, connects to the system bus **706** via the input device interface **740**. In a networked environment, program modules depicted relative to the computer **701**, or portions thereof, can be stored in the remote memory/storage device **746**. It will be appreciated that the network connections shown are exemplary and other means of establishing a communications link between the computers can be used.

(161) The computer **701** is operable to communicate with wire and wireless devices or entities using the IEEE 802 family of standards, such as wireless devices operatively disposed in wireless communication (e.g., IEEE 802.13 over-the-air modulation techniques). This includes at least Wi-Fi (or Wireless Fidelity), WiMax, and Bluetooth™ wireless technologies, among others. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices. Wi-Fi networks use radio technologies called IEEE 802.13x (a, b, g, n, etc.) to provide secure, reliable, fast wireless connectivity. A Wi-Fi network can be used to connect computers to each other, to the Internet, and to wire networks (which use IEEE 802.3-related media and functions).

(162) FIG. **8** is a block diagram depicting an exemplary communications architecture **800** suitable for implementing various embodiments as previously described. The communications architecture **800** includes various common communications elements, such as a transmitter, receiver, transceiver, radio, network interface, baseband processor, antenna, amplifiers, filters, power supplies, and so forth. The embodiments, however, are not limited to implementation by the communications architecture **800**.

(163) As shown in FIG. **8**, the communications architecture **800** includes one or more clients **802** and servers **804**. The clients **802** may implement the client device **510**. The servers **804** may implement the server device **526**. The clients **802** and the servers **804** are operatively connected to one or more respective client data stores **806** and server data stores **808** that can be employed to store information local to the respective clients **802** and servers **804**, such as cookies and/or associated contextual information.

(164) The clients **802** and the servers **804** may communicate information between each other using a communication framework **810**. The communications framework **810** may implement any well-known communications techniques and protocols. The communications framework **810** may be implemented as a packet-switched network (e.g., public networks such as the Internet, private networks such as an enterprise intranet, and so forth), a circuit-switched network (e.g., the public switched telephone network), or a combination of a packet-switched network and a circuit-switched network (with suitable gateways and translators).

(165) The communications framework **810** may implement various network interfaces arranged to accept, communicate, and connect to a communications network. A network interface may be regarded as a specialized form of an input output interface. Network interfaces may employ connection protocols including without limitation direct connect, Ethernet (e.g., thick, thin, twisted

pair 10/100/1000 Base T, and the like), token ring, wireless network interfaces, cellular network interfaces, IEEE 802.8a-x network interfaces, IEEE 802.16 network interfaces, IEEE 802.20 network interfaces, and the like. Further, multiple network interfaces may be used to engage with various communications network types. For example, multiple network interfaces may be employed to allow for the communication over broadcast, multicast, and unicast networks. Should processing requirements dictate a greater amount speed and capacity, distributed network controller architectures may similarly be employed to pool, load balance, and otherwise increase the communicative bandwidth required by clients **802** and the servers **804**. A communications network may be any one and the combination of wired and/or wireless networks including without limitation a direct interconnection, a secured custom connection, a private network (e.g., an enterprise intranet), a public network (e.g., the Internet), a Personal Area Network (PAN), a Local Area Network (LAN), a Metropolitan Area Network (MAN), an Operating Missions as Nodes on the Internet (OMNI), a Wide Area Network (WAN), a wireless network, a cellular network, and other communications networks.

IV. GENERAL NOTES ON TERMINOLOGY

(166) Some embodiments may be described using the expression “one embodiment” or “an embodiment” along with their derivatives. These terms mean that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment. Moreover, unless otherwise noted the features described above are recognized to be usable together in any combination. Thus, any features discussed separately may be employed in combination with each other unless it is noted that the features are incompatible with each other.

(167) With general reference to notations and nomenclature used herein, the detailed descriptions herein may be presented in terms of program procedures executed on a computer or network of computers. These procedural descriptions and representations are used by those skilled in the art to most effectively convey the substance of their work to others skilled in the art.

(168) A procedure is here, and generally, conceived to be a self-consistent sequence of operations leading to a desired result. These operations are those requiring physical manipulations of physical quantities. Usually, though not necessarily, these quantities take the form of electrical, magnetic or optical signals capable of being stored, transferred, combined, compared, and otherwise manipulated. It proves convenient at times, principally for reasons of common usage, to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like. It should be noted, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to those quantities.

(169) Further, the manipulations performed are often referred to in terms, such as adding or comparing, which are commonly associated with mental operations performed by a human operator. No such capability of a human operator is necessary, or desirable in most cases, in any of the operations described herein, which form part of one or more embodiments. Rather, the operations are machine operations. Useful machines for performing operations of various embodiments include general purpose digital computers or similar devices.

(170) Some embodiments may be described using the expression “coupled” and “connected” along with their derivatives. These terms are not necessarily intended as synonyms for each other. For example, some embodiments may be described using the terms “connected” and/or “coupled” to indicate that two or more elements are in direct physical or electrical contact with each other. The term “coupled,” however, may also mean that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other.

(171) Various embodiments also relate to apparatus or systems for performing these operations. This apparatus may be specially constructed for the required purpose or it may comprise a general purpose computer as selectively activated or reconfigured by a computer program stored in the

computer. The procedures presented herein are not inherently related to a particular computer or other apparatus. Various general purpose machines may be used with programs written in accordance with the teachings herein, or it may prove convenient to construct more specialized apparatus to perform the required method steps. The required structure for a variety of these machines will appear from the description given.

(172) It is emphasized that the Abstract of the Disclosure is provided to allow a reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in a single embodiment for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separate embodiment. In the appended claims, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein,” respectively. Moreover, the terms “first,” “second,” “third,” and so forth, are used merely as labels, and are not intended to impose numerical requirements on their objects.

(173) What has been described above includes examples of the disclosed architecture. It is, of course, not possible to describe every conceivable combination of components and/or methodologies, but one of ordinary skill in the art may recognize that many further combinations and permutations are possible. Accordingly, the novel architecture is intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims.

V. CONCLUSION

(174) Any or all of the above-described techniques may be implemented by suitable logic stored on a non-transitory computer-readable medium. When executed by one or more processors, the logic may cause the processors to perform the techniques identified above. The logic may be implemented fully or partially in hardware. The logic may be included as part of a controller for controlling the actuation, de-actuation, movement, position, etc. of a soft robotic actuator and/or a soft robotic system employing one or more actuators in a gripper arrangement.

Claims

1. A system comprising: a display configured to show a control interface; an interface configured to transmit control commands to a pick-and-place station, the pick-and-place station comprising a robotic gripper attached to a robotic arm, at least one source location, and at least one target location; and a processor; and a memory storing instructions configured to: display, on the interface, a view of the source location or target location; capture an image of the source location or the target location at a first time when the robotic arm is not present in the view; identify, at a second time, that the robotic arm is present in the view; and use the captured image on the display at the second time.
2. The system of claim 1, wherein capturing the image comprises capturing images at predetermined intervals.
3. The system of claim 1, wherein capturing the image comprises capturing the image immediately before the robotic arm is commanded to move into the view.
4. The system of claim 1, wherein using the captured image comprises replacing a camera view at the second time with the captured image.
5. The system of claim 1, wherein using the captured image comprises partially superimposing the captured image on a current camera view to create a ghost image.
6. The system of claim 1, the instructions further configured to: receive an actuator sequencing

specifying an order in which different actuators of the robotic gripper are deployed; and transmit a control signal configured to cause the robotic gripper to actuate different actuators in the specified order.

7. The system of claim 1, the instructions further configured to: receive an input gesture on the interface configured to perform a predefined action with the robotic arm or gripper.

8. The system of claim 7, wherein the input gesture comprises one or more of: a flicking gesture configured to cause the robotic gripper to rearrange a target object or nudge the target object; or a stirring gesture configured to cause at least one of the robotic arm or the robotic gripper to stir objects at the source or target location.

9. The system of claim 1, the instructions further configured to: receive an input path that defines the path that the robotic arm should take in order to approach a target object.

10. The system of claim 9, wherein the input path is input by detecting a location of a user's arm using a camera capable of three-dimensional motion tracking.

11. The system of claim 9, wherein the input path is input by receiving input from sensors embedded in a glove worn by a user.

12. A method for displaying a control interface for a robotic pick-and-place station comprising a robotic gripper attached to a robotic arm, at least one source location, and at least one target location, the method comprising: displaying, on the control interface, a view of the source location or target location; capturing an image of the source location or the target location at a first time when the robotic arm is not present in the view; identifying, at a second time, that the robotic arm is present in the view; and using the captured image on the display at the second time.

13. The method of claim 12, wherein capturing the image comprises capturing images at predetermined intervals.

14. The method of claim 12, wherein capturing the image comprises capturing the image immediately before the robotic arm is commanded to move into the view.

15. The method of claim 12, wherein using the captured image comprises replacing a camera view at the second time with the captured image.

16. The method of claim 12, wherein using the captured image comprises partially superimposing the captured image on a current camera view to create a ghost image.

17. The method of claim 12, further comprising: receiving an actuator sequencing specifying an order in which different actuators of the robotic gripper are deployed; and transmitting a control signal configured to cause the robotic gripper to actuate different actuators in the specified order.

18. The method of claim 12, further comprising: receiving an input gesture on the interface configured to perform a predefined action with the robotic arm or gripper, wherein the input gesture comprises one or more of: a flicking gesture configured to cause the robotic gripper to rearrange a target object or nudge the target object; or a stirring gesture configured to cause at least one of the robotic arm or the robotic gripper to stir objects at the source or target location.

19. The method of claim 12, further comprising receiving an input path that defines the path that the robotic arm should take in order to approach a target object.

20. The method of claim 19, wherein the input path is input by at least one of: detecting a location of a user's arm using a camera capable of three-dimensional motion tracking, or receiving input from sensors embedded in a glove worn by a user.
